

Product Requirements Specification (PRS)

LIVEFAKENEWS – Reddit Gaming Fact-Checking System

1. Product Overview

Product Name: LIVEFAKENEWS

Description: LIVEFAKENEWS is an AI-powered fact-checking system built specifically to verify gaming-related claims posted on Reddit. It retrieves external evidence, performs semantic evaluation, and produces a verdict of True, False, or Unverified along with supporting citations.

Target Users: Reddit users, moderators, gaming communities, researchers, and content creators.

Value Proposition: Prevents misinformation spread in gaming subreddits by giving users real-time, evidence-based verification in an automated, transparent manner.

2. Problem Statement

Gaming communities on Reddit frequently face issues related to misinformation. Rumors about game release dates, fake patch notes, leaked gameplay screenshots, or fabricated announcements spread rapidly. Since gaming audiences engage heavily with speculative content, false claims tend to go viral quickly. Moderators cannot manually verify the authenticity of every post, especially in high-traffic subreddits.

Existing tools are insufficient because they:

- Only detect linguistic patterns without validating external facts.
- Are not optimized for gaming terminology or context.
- Cannot handle fast-moving discussions where corrections come too late.

LIVEFAKENEWS addresses these gaps by combining claim extraction, real-time web search, semantic evidence retrieval, and transformer-based verification to determine the authenticity of statements with transparency and reliability.

3. Objectives

- Provide automated, real-time claim verification for Reddit gaming content.
- Retrieve trustworthy information from credible web sources.
- Deliver explanations with evidence and citations for user trust.
- Be resilient to paraphrasing, slang, and low-quality writing.
- Achieve stable performance: high accuracy with low latency.

Success Criteria:

- $\geq 83\%$ F1-macro score for classification.
- ≤ 3 seconds latency for complete fact-check.
- Clear reasoning and citations for every output.
- User satisfaction $\geq 85\%$ in user study.

4. Scope

In Scope:

Extraction of factual claims from Reddit posts.

- Real-time Google search via SerpAPI.
- Web scraping and text extraction.
- Semantic embedding and FAISS-based retrieval.
- Cross-encoder verification of evidence.
- Domain classifier for rumor-likelihood scoring.
- Aggregated decision-making and FastAPI integration.

Out of Scope:

- Fact-checking political, medical, or financial claims (Phase 1).
- Moderation tools requiring human-in-the-loop review.
- Non-Reddit platforms (Twitter, Discord, etc.).

5. Core Features

Claim Extraction: Identifies a concise, verifiable claim from the post's title and body.

Web Evidence Retrieval: Uses Google search to locate relevant articles, posts, and announcements.

Semantic Retrieval: MiniLM embeddings + FAISS index enable vector-based similarity search for evidence.

Cross-Encoder Reranking: Compares claim–evidence pairs to evaluate truth likelihood.

Domain Classifier: DeBERTa-based classifier trained on gaming misinformation datasets.

Explanation Generator: Produces human-readable reasoning with citation links.

6. Data Requirements

Training Data: 48k labeled gaming-domain texts used for domain classifier tuning.

Evidence Data: Dynamically retrieved via SerpAPI and scraped webpages.

Model Data: Pretrained sentence transformers, cross-encoders.

Privacy Considerations:

- No user-identifiable data is stored.
- Scraping only permitted public information.
- Data processing follows Reddit and SerpAPI policies.

7. System Behavior

Inputs:

- Reddit post title
- Reddit post body (optional)

Outputs:

- Final classification: True / False / Unverified
- Explanation summarizing evidence
- Evidence citations (top 3 URLs)
- Domain classifier confidence
- Retrieval and reranking scores

Performance Requirements:

- Full pipeline response: ≤ 3 seconds
- Cross-encoder inference optimized for GPU
- Minimum 99% system uptime

8. Constraints

Technical:

- SerpAPI query limits
- GPU requirement for real-time verification
- Scraper failures due to blocked URLs

Regulatory:

- Must follow robots.txt rules for scraping
- Must respect Reddit API rate limits

Ethical:

- Avoid mislabeling due to incomplete evidence
- Ensure fairness across all types of gaming content

9. Metrics & Evaluation

Quantitative Metrics:

- Accuracy, Precision, Recall, F1 Score
- Evidence relevance score
- Latency (per request)

Qualitative Metrics:

- User trust and satisfaction
- Moderator feedback

Target Thresholds:

- $F1 \geq 83\%$
- Latency ≤ 3 seconds
- User approval $\geq 80\%$

10. Implementation Notes

Deployment uses FastAPI, hosted on Render/GCP/Railway. The system performs runtime retrieval, on-device embedding, FAISS search, cross-encoder inference, and final aggregation. Caching frequently used sources can reduce search overhead.

11. Timeline & Priorities

Phase 1: Core pipeline (extraction → search → scraping → retrieval → reranking → aggregation).

Phase 2: Evidence caching, interactive dashboard, Reddit bot deployment.

Phase 3: Multi-domain fact-checking, LLM-enhanced reasoning, mobile/web UI.

12. Open Questions

- Should credibility scores for sources be included?
- Should contradictory evidence be flagged separately?
- Should multilingual posts be handled in Phase 2?
- Should system output user-friendly summaries or detailed reports